CC LAB 8

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SECTION: B

```
24-03-2026  14:54    <DIR>          .
24-03-2026  14:55    <DIR>          ..
24-03-2026  14:54               562 docker-compose.yml
24-03-2026  14:54               450 Dockerfile
24-03-2026  14:54             1,377 node.py
24-03-2026  14:54                 6 requirements.txt
               4 File(s)          2,395 bytes
               2 Dir(s)  412,950,421,504 bytes free
```

```
#21 [node2] resolving provenance for metadata file
#21 DONE 0.0s
[+] up 7/7
 ✔Image raft_docker_lab-node1      Built
 ✔Image raft_docker_lab-node2      Built
 ✔Image raft_docker_lab-node3      Built
 ✔Network raft_docker_lab_raftnet Created
 ✔Container node1                  Created
 ✔Container node2                  Created
 ✔Container node3                  Created
```

```
node1 | [FOLLOWER Node 1] Current leader: 172.28.0.3:8002
node2 | [LEADER Node 2] I am the leader
node3 | [FOLLOWER Node 3] Current leader: 172.28.0.3:8002
```

```
Starting Node 1 at 172.28.0.2:8001
[Node 1] Waiting for leader election...
[Node 1] Leader changed: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
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[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
```

```
Starting Node 2 at 172.28.0.3:8002
[Node 2] Waiting for leader election...
[Node 2] Leader changed: 172.28.0.3:8002
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
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[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
```

```
Starting Node 3 at 172.28.0.4:8003
[Node 3] Waiting for leader election...
[Node 3] Leader changed: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
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[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
e6577990d4b7	raft_docker_lab-node2	"python node.py 2"	8 minutes ago	Up 8 minutes		node2
a3629306023f	raft_docker_lab-node3	"python node.py 3"	8 minutes ago	Up 8 minutes		node3

```
[
    {
        "Name": "raft_docker_lab_raftnet",
        "Id": "aa0d11da2190e824e0f971dc71cd1bf22520be4fcf431a3484a645fd71cfe7a0",
        "Created": "2026-03-24T09:30:06.805298677Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": [
                {
                    "Subnet": "172.28.0.0/16",
                    "Gateway": "172.28.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {
            "com.docker.network.enable_ipv4": "true",
            "com.docker.network.enable_ipv6": "false"
        },
        "Labels": {
            "com.docker.compose.config-hash": "7b95512846367755 9d4edba13e9fe8f69dd803b2dfc190dd0f2333946fc6a5d0",
            "com.docker.compose.network": "raftnet",
            "com.docker.compose.project": "raft_docker_lab",
            "com.docker.compose.version": "5.1.0"
        },
        "Containers": {
            "a3629306023fc6c9e1357001ad899a1c153a84345ff71b64f3946aaf8dddf564": {
                "Name": "node3",
                "EndpointID": "aef6c997dfdded3dcab76a36f0e6e7f3d004aebb59c6edf1d235485883809ed8",
                "MacAddress": "7e:a7:52:e3:b2:cb",
                "IPv4Address": "172.28.0.4/16",
                "IPv6Address": ""
            },
            "a7b35ebde558ae860a26ba985ea9882afe29e1876f79e1fd84e7a6b3ef6ef2f1": {
                "Name": "node1",
                "EndpointID": "04c82e5d6be01c5d9f989aaa15351df2f0c2565fb581cd3febb5b21b64c6fbb7",
                "MacAddress": "1a:28:25:26:01:6f",
                "IPv4Address": "172.28.0.2/16",
                "IPv6Address": ""
            },
            "e6577990d4b77b45f114cf3d5ea7b683d3b294db455aa03de6c4a0188b793b36": {
                "Name": "node2",
                "EndpointID": "9bd48e7c8951788e25067351e44fe6617b45a1d9c8950c52a0405946 4f9f8927",
                "MacAddress": "02:5e:08:81:f3:be",
                "IPv4Address": "172.28.0.3/16",
                "IPv6Address": ""
            }
        },
        "Status": {
            "IPAM": {
                "Subnets": {
                    "172.28.0.0/16": {
                        "IPsInUse": 6,
                        "DynamicIPsAvailable": 65530
                    }
                }
            }
        }
```


```
node1   |  [FOLLOWER Node 1] Current leader: 172.28.0.4:8003
node3   |  [LEADER Node 3] I am the leader
node1   |  [FOLLOWER Node 1] Current leader: 172.28.0.4:8003
node3   |  [LEADER Node 3] I am the leader
```

```
[Node 1] Leader changed: None
[Node 1] Waiting for leader election...
[Node 1] Leader changed: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
```

```
[Node 3] Leader changed: None
[Node 3] Waiting for leader election...
[Node 3] Leader changed: 172.28.0.4:8003
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
```

```
node2 | [FOLLOWER Node 2] Current leader: 172.28.0.4:8003
node1 | [FOLLOWER Node 1] Current leader: 172.28.0.4:8003
node3 | [LEADER Node 3] I am the leader
```

```
[Node 2] Leader changed: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
```

Activity

5 nodes

```
CONTAINER ID   IMAGE                    COMMAND                  CREATED          STATUS          PORTS     NAMES
e053f078de1f   raft_docker_lab-node5    "python node.py 5"       33 seconds ago   Up 32 seconds             node5
2945d894a550   raft_docker_lab-node2    "python node.py 2"       33 seconds ago   Up 32 seconds             node2
3381d21a293d   raft_docker_lab-node1    "python node.py 1"       33 seconds ago   Up 32 seconds             node1
4ea60c7637f1   raft_docker_lab-node3    "python node.py 3"       33 seconds ago   Up 32 seconds             node3
f8832e7bbd7c   raft_docker_lab-node4    "python node.py 4"       33 seconds ago   Up 32 seconds             node4
```

```
Starting Node 4 at 172.28.0.5:8004
[Node 4] Waiting for leader election...
[Node 4] Leader changed: 172.28.0.5:8004
[LEADER Node 4] I am the leader
[LEADER Node 4] I am the leader
[LEADER Node 4] I am the leader
[LEADER Node 4] I am the leader
[LEADER Node 4] I am the leader
[LEADER Node 4] I am the leader
[LEADER Node 4] I am the leader
[LEADER Node 4] I am the leader
[LEADER Node 4] I am the leader
```

```
Starting Node 3 at 172.28.0.4:8003
[Node 3] Waiting for leader election...
[Node 3] Leader changed: 172.28.0.5:8004
[FOLLOWER Node 3] Current leader: 172.28.0.5:8004
[FOLLOWER Node 3] Current leader: 172.28.0.5:8004
[FOLLOWER Node 3] Current leader: 172.28.0.5:8004
[FOLLOWER Node 3] Current leader: 172.28.0.5:8004
[FOLLOWER Node 3] Current leader: 172.28.0.5:8004
[FOLLOWER Node 3] Current leader: 172.28.0.5:8004
[FOLLOWER Node 3] Current leader: 172.28.0.5:8004
[FOLLOWER Node 3] Current leader: 172.28.0.5:8004
[FOLLOWER Node 3] Current leader: 172.28.0.5:8004
```

```
Starting Node 2 at 172.28.0.3:8002
[Node 2] Waiting for leader election...
[Node 2] Leader changed: 172.28.0.5:8004
[FOLLOWER Node 2] Current leader: 172.28.0.5:8004
[FOLLOWER Node 2] Current leader: 172.28.0.5:8004
[FOLLOWER Node 2] Current leader: 172.28.0.5:8004
[FOLLOWER Node 2] Current leader: 172.28.0.5:8004
[FOLLOWER Node 2] Current leader: 172.28.0.5:8004
[FOLLOWER Node 2] Current leader: 172.28.0.5:8004
[FOLLOWER Node 2] Current leader: 172.28.0.5:8004
[FOLLOWER Node 2] Current leader: 172.28.0.5:8004
```

```
Starting Node 1 at 172.28.0.2:8001
[Node 1] Waiting for leader election...
[Node 1] Leader changed: 172.28.0.5:8004
[FOLLOWER Node 1] Current leader: 172.28.0.5:8004
[FOLLOWER Node 1] Current leader: 172.28.0.5:8004
[FOLLOWER Node 1] Current leader: 172.28.0.5:8004
[FOLLOWER Node 1] Current leader: 172.28.0.5:8004
[FOLLOWER Node 1] Current leader: 172.28.0.5:8004
[FOLLOWER Node 1] Current leader: 172.28.0.5:8004
[FOLLOWER Node 1] Current leader: 172.28.0.5:8004
[FOLLOWER Node 1] Current leader: 172.28.0.5:8004
[FOLLOWER Node 1] Current leader: 172.28.0.5:8004
[FOLLOWER Node 1] Current leader: 172.28.0.5:8004
```

```
Starting Node 5 at 172.28.0.6:8005
[Node 5] Waiting for leader election...
[Node 5] Leader changed: 172.28.0.5:8004
[FOLLOWER Node 5] Current leader: 172.28.0.5:8004
[FOLLOWER Node 5] Current leader: 172.28.0.5:8004
[FOLLOWER Node 5] Current leader: 172.28.0.5:8004
[FOLLOWER Node 5] Current leader: 172.28.0.5:8004
[FOLLOWER Node 5] Current leader: 172.28.0.5:8004
[FOLLOWER Node 5] Current leader: 172.28.0.5:8004
[FOLLOWER Node 5] Current leader: 172.28.0.5:8004
[FOLLOWER Node 5] Current leader: 172.28.0.5:8004
[FOLLOWER Node 5] Current leader: 172.28.0.5:8004
[FOLLOWER Node 5] Current leader: 172.28.0.5:8004
[FOLLOWER Node 5] Current leader: 172.28.0.5:8004
[FOLLOWER Node 5] Current leader: 172.28.0.5:8004
```

```
C:\Users\asmi__>docker stop node1
node1

C:\Users\asmi__>docker stop node2
node2

C:\Users\asmi__>docker ps
CONTAINER ID   IMAGE                   COMMAND              CREATED         STATUS         PORTS     NAMES
e053f078de1f   raft_docker_lab-node5   "python node.py 5"   5 minutes ago   Up 5 minutes             node5
4ea60c7637f1   raft_docker_lab-node3   "python node.py 3"   5 minutes ago   Up 5 minutes             node3
f8832e7bbd7c   raft_docker_lab-node4   "python node.py 4"   5 minutes ago   Up 5 minutes             node4
```

```
C:\Users\asmi__>docker stop node3
node3

C:\Users\akans>docker ps
CONTAINER ID   IMAGE                   COMMAND              CREATED         STATUS         PORTS     NAMES
e053f078de1f   raft_docker_lab-node5   "python node.py 5"   8 minutes ago   Up 8 minutes             node5
```

```
[Node 4] Leader changed: None
[Node 4] Waiting for leader election...
[Node 4] Waiting for leader election...
[Node 4] Waiting for leader election...
[Node 4] Waiting for leader election...
[Node 4] Waiting for leader election...
```

```
[Node 5] Leader changed: None
[Node 5] Waiting for leader election...
[Node 5] Waiting for leader election...
[Node 5] Waiting for leader election...
[Node 5] Waiting for leader election...
[Node 5] Waiting for leader election...
[Node 5] Waiting for leader election...
```

Q: Why did leader election not occur when only 2 nodes were active in a 5- node cluster?

Answer: Leader election did not occur because Raft requires a majority (quorum) of nodes to elect a leader.
In a 5-node cluster, the majority is calculated using:

(N / 2) + 1

For 5 nodes:

(5 / 2) + 1 = 3

So at least 3 nodes must be active to form a quorum.

When only 2 nodes are active, they cannot reach the required majority (3 votes). No candidate can receive enough votes. Therefore, no leader is elected. This prevents a split- brain
problem and ensures consistency in the distributed system.